

ActInf GuestStream #014.1 ~ Osteopathy and Mental Health (Bohlen, Esteves, Cerritelli, Shaw)

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hello and welcome everyone to the active inference lab today it is january 10 2022 and we're here in active guest stream number 14 with a whole host of awesome guests we're going to be discussing osteopathy and mental health and we're going to have the authors say hello and introduce themselves then lucas will give a presentation and during the presentation please feel free to add your comments in questions in the live chat and then we'll have a bunch of time after the presentation to ask those questions to the authors hear any author's remarks so really excited to hear from you all and learn from your perspective let's hear the introductions from the authors and then go to lucas so please robert and then we'll kick it off from there thank you uh yeah robert shaw i'm based in sweden i'm an osteopath and a psychotherapist my research interest is particularly with embodiment and that was what my phd was on embodiment within the therapeutic relationship hi everyone i'm francesco italy from italy uh i'm a neuroscientist and an osteopath uh mainly um interest in uh discovering the uh how the brain works uh in relation to touch in relation to the effect of touch based on the different type of stimuli that the the body received hi uh i'm george ester jestervilles i'm an osteopath i'm currently based in malta and italy i'm also a cognitive scientist and my main interest is on predictive processing active entrance touch and chronic pain in general yeah hi my name is uh lucas i think i'm the last of the authors so i'm an osteopath as well based in hamburg in germany and i'm currently working part-time at the osteopathic research institute and i'm working there in education education and research and i'm also part-time studying psychology and neuroscience at mentor augmented health at king's college in london and i think i can kick off the presentation now that's okay with you tanya perfect thanks so um yeah we have uh the opportunity thanks again for the active appearance lab for the invitation it's it's a pleasure to be here we have the opportunity to present a paper that we have recently published in frontiers in psychology in the section health psychology the research topic called an activism and active appearance in the therapeutic alliance and the paper is entitled osteopathy and mental health

and embodied predictive and introspective framework and um yeah without further ado i think let's let's dive right into it um a little introduction on why these things might be interesting um we have a globally a very high prevalence and a burden but also comorbidity between musculoskeletal and uh mental health disorders so to put this in perspective about uh one billion prevalent cases of mental disorders have been reported in 2017 while about 1.3 billion prevalent cases have been reported of musculoskeletal disorders in 2017. so together these uh two broad spectrum of disorders are um very prevalent but also burdensome so together both both conditions contribute um to other main contributors to to work by disability but um musculoskeletal mental disorders also highly co-morbid which means for example that anxiety depression are frequently core core with chronic and often musculoskeletal pain conditions and vice versa for example depressive symptoms are quite prevalent and pain suffers and pain symptoms are also quite prevalent in individuals with depression um so these uh this data kind of outlines in a need to to integrate physical and mental health care services in our opinion but also in the opinion of many research papers in the literature so um especially multidisciplinary collaborations between most skeleton and psychological healthcare specialists have been called for but on the other hand there have also been some disease classifications that have been proposed to to combine physical and mental health and propose classifications that are neither purely somatic nor julian mental we believe that there is this need to integrate these kind of approaches and we like to propose osteopathy to be to be one of these approaches and osteopathy for those who don't know is generally used primarily to treat musculoskeletal disorders like back pain but um it is also uh has recently been proposed to benefit psychosocial factors in particular in in patient populations with persistent pain there have also there was also some research that has been conducted from hillary abbey and colleagues showing some positive outcomes of combined treatment as a combined osteopathy and combined psychological informed strategies uh showing positive results so there are only a few attempts to to investigate the effectiveness of osteopathy in the field of mental health and there are only a few approaches being developed to treat or to engage with mental health based on this background we proposed um we tried to propose a framework on how to move forward and how to research the putative effects of osteopathy in this field and we do so by drawing together um from the research fields of embodied cognition predictive coding but also interception and osteopathy and we like to propose in the end our framework which is an embodied predictive and introspective remote and if you look on the right hand side figure one this is kind of our background put together and to simplify body cognition kind of highlights

the role of the body and cognition and the mental health and these process processes are likely underpinned by predictive coding processes furthermore our access point we discuss interception as an access point and therapeutic access point to to affect these processes and we argue that osteopathy might might be an approach to to realize this aim so let's have a look at the background we start with the first research field which is embodied cognition generally embodied embodiment is an interdisciplinary field of research which spans disciplines including philosophy but also psychology psychiatry and neuroscience and in general these theories of embodiment argue that the cognition and emotion are both dependent on embedded embodied simulations they argue that body and mind are inseparable in producing cognition and that the non-neural body might both constrain and enable cognition but notably there are stronger and weaker versions of embodiment some argue that the body might contribute to cognitive processes while other versions argue that it might constitute cognitive processes so to get a little bit more detail involved theories of embodiment argue that cognition and emotion are based upon reinstatements of perception so this is relating to both external extra receptive and inter meaning interceptive sensory states but also to action which relates to proprioceptive motor states and these perceptions produce embodied or sensory motor simulations which of previous experiences in one's self so this is a kind of explanation of embodiment this perspective is quite interesting because it might revise our brain-centered view of cognition by acknowledging that these processes are not simply limited to neural events in the brain but also cut across the brain body and world divisions and should be also studied accordingly and there are different kinds of commitment to the embodied theories and and activism is an interesting aspect or linked concept that i want to to go into detail a little bit further and an activism proposes that cognition emerges from the dynamic interaction of brain body and also environment so um most of you might have heard about the 4e cognition um which is that cognition is all together embodied embedded and acted extended but also ecological and um we are or at this argument that cognition is an embodied embedded enacted extended and process of sense making through a body in an environment so this is an evaluate evaluative interaction of an organism and its environment so in detail these four or five e's mean that body cognition is enabled through and also constrained by the non-neural body embedded means that cognition depends on the environmental context as the organism is situated within the environment and enacted means that cognition is for action and depends on the interaction of the embodied organism with the and extended means that cognition extends beyond the brain and body into the environment the most notable example might be your

smartphone and lastly ecological means that cognition depends on the environmental affordances for action and all together this might constitute an activism with a 5e perspective so combining these perspectives of embodied and active inactivism both mental and physical health conditions for example pain and um are associated with alterate interaction with an altered interaction of the organism within with with the environment through its embodiment so it's the three-part interaction of brain body and environment so to sum this up embodiment and activism emphasized the interaction of brain body and environment in understanding the mind and cognition to understand the the processes of the brain in a little bit more detail we have to turn to another field of research which is the next part of the background this is predictive coding and i believe that predictive coding is a topic most of the listeners are more familiar with than us but i try to summarize them anyway so predictive processing is a theoretical framework with growing influence in the field of cognitive science and it is closely related but also distant uh distinct from the free energy principle asian brain hypothesis and active inference so i will summarize all of them closely shortly the free energy principle proposes that living systems resist the tendency to disorder or another words remain in thermodynamic non-equilibrium steady states and they do so by restricting themselves to a limited number of states through the minimization of free energy and in in this instance free energy is defined as the difference between a system's predicted and actual states and this can be minimized using both perception and action and perception it is minimized by updating the prediction that is made from the generated model based on the sensation and action in turn is changing the sensation through action to match with the prediction so it's like a self-fulfilling prophecy and the bayesian brain hypothesis proposes that brains make predictions about the causes of extra receptive interceptive but also proprioceptive sensations um using both the current sensory input but also prior beliefs based on the generated model and in general the confidence or precision one has uh in the belief or the prior and the sensory input or the likelihood will determine how much the perception will in the end shift towards the expectation in general a gap between prior and likelihood is called a prediction error or free energy while a system is minimizing prediction error or free energy or surprise or entropy that is also maximizing the evidence for its own model of the world and this is the bargaining brain hypothesis is relating to behavior so predictive coding is putting this into a neural perspective and proposes that there are descending top-down predictions that are conveyed from higher cortical levels down to lower cortical levels where they are compared to us and bottom-up sensory information from this perspective information only goes up the cortical hierarchy and

if a mismatch between the predicted and actual information occurs which is terms of prediction error and otherwise top-down predictions constantly explain away bottom sensory information then there's active imperance which proposes that the brain actively modifies its input to confirm prior expectations and this is an inactive and also selective sampling of sensory information in the world through action prediction errors can be minimized we've talked about it a little bit through perceptual influence and active endurance uh perceptive appearance meaning revising the generated model based on the prediction errors and active inference acting on the world to to generate a state that is predicted by the generative one and to bring all of this together within the physical mental health condition to understand the physical and mental health conditions it might be an altered weighting of prior beliefs and sensory information which gives rise to physical mental so for example physical health conditions prior beliefs are likely overweighted relative to sensory information which means i follow too much prediction uh precision which then predicts symptoms consequently experienced this might also lead to heightened prediction of pain even towards harmless sensations so that harmless sensations might be perceived as painful and on the other hand mental health um both prior beliefs might be overly precise compare in comparison to sensory information or vice versa and so the first one is depression and the second one in psychosis yes one point that was quite interesting is that chronic pain and depression which we are quite interested in are particularly linked to false experiences of interception there's a little literature about this so that's this is our next research topic that we want to uh to combine with the other ones um so what is interception um kaiser and i have defined interception as the process by which the nervous system senses interprets and integrates signals originating from within the body so it's all information from within our own body and this provides uh some kind of embodied sensory experience um which is absolutely necessary for the adaptive interaction with the an interesting aspect is that the active interference model was also applied to interception which is termed interceptive appearance and it proposes that interceptive experiences result from interferences about the hidden causes of this sensory information so to put this into the neural perspective uh top-down interceptive predictions are compared with bottom-up interceptive information most likely in the insula and the mismatch between both then results in a prediction error which is minimized depending on the on the precision by either perceptual inference or active interference so receptor inference means revising the top-down interceptive predictions and active impedance means then modifying the bottom-up interceptive information to convey with a prediction and interestingly uh annual

seth has proposed that an emotion can also be reviewed as interceptive influence because interceptive prediction errors are used to and fear emotional states connect this to physical and mental health conditions um deficits in interceptive processing for example uh low interceptive accuracy um might be associated with both physical and mental health conditions like chronic pain and comorbid um however there is it is not clear how that every disorder in every individual might behave the same in this way i think there are differences so so some disorders in individuals would likely benefit from reducing overly precise interceptive predictions for example robbing beliefs about real but indeed harmless sensations while other patients might might reduce overly precise and receptive information which is for example utilitary sensations that are maintained that maintain robbing beliefs so um there are both possibilities so let's talk about our approach on uh how to influence to interceptive information and this is osteopathy we are all osteopaths uh the authors on this paper so a few words about osteopathy osteopathy combines hands-on manual approaches with hands-off patient management approaches so we use touch and manipulation as hands-on which is informed of sympathetic models of care so our clinical reasoning and hands-off patient-managed approaches involve patient education psychological support lifestyle advice but also self-management solutions and these are informed of course by osteopathic principles but notably both might utilize top-down and bottom-up dynamics between peripheral tissues and the brain in general osteopathy is applied to treat a range of clinical conditions but an emphasis is given on chronic pain conditions in particular musculoskeletal disorders like back pain philosophically osteopathy has emphasized the unity and interaction of body mind and soul for quite some time and it has also been proposed that it might benefit psychological outcomes especially in chronic pain patients but this needs some further research and um francesco and colleagues also proposed that uh osteopathic treatment might influence interceptive processing in a concept paper they published in i think and i went on to to conduct some primary research showing that um osteopathy indeed increases intrceptive accuracy but also decreases uh the bulge response of brain correlates of interception in patients with chronic low back pain um they do so probably by the stimulation of afferent c tactile fibers which communicate interceptive and effective dimensions of touch so this could be relevant to both that involve interceptive difficulties which are for example chronic pain and also depression so this is our background and i think it leads up to to the framework we like to present and the framework bridges all of these fields of research and it is an embodied interceptive but also predictive framework the basic basis is that both physical and mental health symptoms may result from

altered precision waiting between interceptive predictions and intercepted input so we argue that literature argues that um either too much precision is further to the prior which is an overly precise prediction on um or too little precision is referred to the likelihood which is an imprecise prediction error arguably chronic pain and depression are mainly linked to the overweighting of priors so um these beliefs likely predict painful and depressive states even if the actual interceptive information is harmless and they do so due to past experiences and they likely make these states come true through active interference so the expected symptoms might be generated through action to confirm the prediction and in order to update these prior release and reduce the active influence of pain and and depressed states it is likely that surprising interceptive input needs to be provided and to increase first of all the weight of the likelihood but also to generate prediction errors that can revise the belief issuing the predictions and thus osteopaths likely need to foster perceptual inference so actual information is used in perceptual inference to update the prediction and also the perception of these symptoms and we have a look at figure two in a moment so um this uh there's a paper by far from 2015 which which has argued that contemplative practices may alter interceptive processing by shifting regulatory habits from active to perceptual inference and i will we believe that that osteopathy does the same if we have a look at our main figure here um we propose that we see a patient in red lying on the treatment table imagine this patient has um chronic pain with comorbid depression and he's lying there and the osteopath provides some touch-based interventions to the symptomatic area in this this image the abdomen we argue that the belief or the prayer of the patient is likely to predict physical and mental states that are associated with pain and depression because uh to be the likely likely cause for uncertain information this is the the overweighting of the prior if we provide some treatment interventions to the patient the sensory input is not linked to these physically mental states which is surprising but osteopathic treatment often does not involve pain but is quite pleasant if you if you touch a patient so this is a surprising mismatch of course to the patients to the patient because the expected and the actual information do not match thus generate an interceptive prediction error so these addition errors need to be minimized and they can be minimized using active or perceptual inference and in chronic pain we argue that high precision is afforded to the belief to the prior and low precision is further to the sensory information and the likelihood so the first process that needs to be engaged is active interference um which likely produces the symptoms uh that are resembling the predicted states so likely the autonomic nervous system is engaged to produce sensations that resemble

the expected painful and depressive feeling so uh however we are in a healthcare context in osteopathic treatment so the patient might also be declining uh declined to uh to think that this is a health promoting intervention so um this is not not uh this is contradictory with the uh with a prediction so um it's active influence might not be sufficiently to to reduce and explain the interceptive prediction errors so consequently we argue that perceptual appearances processes might be engaged um to update then the belief and the prior based on the likelihood so the sensory information provided posted by the osteopath might update the belief by the patient and thereby revising the generative model holding the belief and issuing the um we argue that these are the processes that underpin osteopathic treatment um in these kind of patients and especially when an emotional reaction and treatment develops yeah and in essence we argue that uh belief and the prediction of physical depression are updated through these interventions so the the persistent and noisy interceptive additions errors which maintain these symptoms through active interference might be replaced with surprising and precise interceptive prediction errors to alleviate the symptoms through receptor computers so this is in a nutshell our framework and we also of course need to discuss this framework and have a look at some implications also limitations so in general i like to emphasize that this theoretical framework is a theoretical framework and therefore requires experimental scrutiny for verification or falsification so future research particularly needs to explore if osteopathic treatment indeed alleviates physical and mental symptoms in this patient population also if it actually increases interceptive accuracy in this population and likely also decreases allostatic load and modulates autonomic nervous system activity so there are a few measurement tools we can use for that and um other limitations include the disregard of complexity both of the health conditions but also of the therapeutic approaches so osteopathy and mental and depression and chronic pain are all very complex conditions and we have simplified them like extremely so we overlooked environmental factors and we didn't go into explaining the patient therapist relationship in detail as well but this framework should be understood in this context so it should be understood in a person-centered context and future studies might also have a look at why manual treatment of peripheral tissues sometimes leads to autonomic and emotional responses and i think this is the basic of why i was so interested in this research and these theories because in practice that all frequently engaged patients which responded quite emotionally to touch and um fake quite better after so i wanted to to understand how this might happen so based on our hypothesis you can see in figure 3 panel 8 the therapeutic input that is provided during treatment may contradict the predicted state of pain and and produce an

interceptive prediction which is used to infer emotional states and need to minimize minimize by active influence harvest perceptive and what you can see in nla is that the initial active influence is quite quick and sharp and we have seen or i've seen my colleagues as well from experience that often if patients display emotional responses during treatment they firstly experienced some sympathetic activity like sweating and muscle tightness combined with emotional distress for example starting to cry but these initial reactions frequently are replaced with very parasympathetic activity so the breathing goes down and the patient starts to relax and emotional calmness starts to set in and i believe this is quite well explainable with the framework we have provided here because an active interference might be the first way to engage in minimizing these prediction errors and perceptive influence might be the second and more sustainable way of minimizing these predictions lastly you can see on panel b another idea which might come from this framework and this is the ground for multidisciplinary collaborations in the treatment of mental and physical disorders so we propose a kind of integrative interceptive exposure therapy in which both physical but also mental health specialists can come together and work in a multi-disciplinary but also person-centered care approach and the idea is if you can see on the left-hand side here is that mindfulness practices likely work top down uh helping the patient to attend to the sensation that he or she is experiencing um but also affecting the sensory information like osteopathy working bottom up also affecting the sensory information by likely producing a sensation itself and psychotherapy is slightly toward the top down to help identify and reflect on the sensation and likely works on works on the prior the belief directly and not through the sensory so um these are a few future research ideas and now we can conclude this presentation and to put it all together this hypothesis and theory article proposes the embodied predictive and interests of the framework and it does so to reason and research the effect of osteopathic treatment um hesitatively on comorbid psychological factors in patients with persistent physical conditions one of the basic ideas is that persistent physical and mental conditions like chronic pain and may be underlined by first overly precise interceptive predictions meaning the symptom congruent information is expected and second imprecise interceptive prediction errors meaning that symptom incongruent information is not sensed so osteopathic treatment may provide uncertain and surprising interceptive information to the bodily area that is associated with the physical and also comorbid mental symptoms resulting in interceptive prediction errors that first engage active impedance processes which lead to initial autonomic activity resembling the predicted symptoms and this is because it is the most likely cause of uncertain interceptive information

due to the past experiences of the patient and afterwards due to the contextual factors uh the patient likely engages perceptual appearance processes that then update the maladaptive intercept predictions based on the actual interceptive information and positivity might also actually improve the ability to attend to actual intercepted information so in a nutshell persistent and noisy maintain the symptoms are likely replaced with surprising and precise and receptive prediction errors that arguably alleviate the sentence and in this way treatment might reduce the belief about and also the prediction of physical emitted states that are associated with chronic pain and comorbid depression and we are currently also at the conducting research into these approaches so thorson has also joined us today um has developed an approach to psychosomatic osteopathy and we are currently having a look on how these techniques might influence stress levels and healthy participants but also patients with chronic attention type headaches so chronic musculoskeletal disorders so maybe there's some things to follow in the future so that's it thank you very much for your attention i'm really looking forward to the discussion and to engaging uh robert george and francesco um and here are the references i think i scrolled through them shortly and if someone's interested i think they can get the presentation said right daniel so that's it thanks lucas that was an awesome presentation really a lot to think and talk about perhaps we'll first just open it up to any of the authors and colleagues here please just uh jump in wherever you'd like i think it'd be awesome to hear about just your overview or what brought you to this area of research as well as anything that you picked up on from this last presentation i think maybe i can say a few words because i forgot something in the beginning so um it all started with with george and francesco um inviting inviting uh robert and me to join a research group that um tries to to reconceptualize osteopathic care under the active endurance framework so we have a group of osteopaths from europe working with the influence framework and there are also other papers coming in a special topic which is also uh has been added by edited by george and francesco so there's there are other ideas coming forward in the near future i think you know one one thing that um i guess brought us into this area you know it's a sort of a way of looking into how we can best help our patients in particularly those with um you know present with complexity um including you know chronic pain but not just running by uh you know the the broad sort of uh area of persistent physical symptoms you know those that tend to sort of um go around see different practitioners um look for a solution to their problem try to know what's going on and you know we tend to sort of seat them in practice and struggle at times if we use um you know a very reductionist approach to practice which is do a manual intervention and hope for the

best you know that's that's far from um you know from what is best practice this day is what this patient-centered care and also can you still here i just lost george for an audio i was just gonna ask me too okay i i can pick up a bit [Music] there's been a kind of a problem it with osteopathic theory over the years um which george is uh hinting at around the reductionistic the bio biomechanical model and when it comes to complexity those sorts of traditional ways of looking at people's chronic symptoms um leave something to be well they they they're not really really things missing we we're realizing using uh if we draw from other disciplines that we can see that there's a potentially much better way to describe why people have these kind of chronic type symptoms and um i i came into this a bit like lucas with um being quite surprised in practice about how people reacted to treatment and i think we have this um it's almost a kind of a tacit uh sort of thing within the profession that we we treat people with emotional disturbance but we don't actually up front it and so this conceptual framework is actually starting to redress the balance um which i think is really really important and trying to understand why is it people react um i think the this notion of prediction areas is um kind of an interesting one i i came into it a long time ago looking at it from a psychodynamic point of view and things called ab reactions but i think what we're moving towards is trying to come up with a bit more of a unified theory and that's why i think concepts of active inference and activism are actually starting to help and also the participatory sense making a movement also adds a lot to how we can start to understand it i know luke has mentioned we haven't talked about the therapeutic relationship but we are there are other papers in the pipeline that we'll look at i'm losing my voice i hope that sort of did a bit of a summary for you george i mean very interesting george no no no i i think robert you know did a brilliant job in summarizing everything so you know all the challenges and the reasons why you know we we are interested in into exploring a different kind of way of of looking at things francesca or torsten would you like to add anything um otherwise anyone in the live chat is welcome to add a question and i have some questions too uh yes daniel so and one other um viewpoint that i cannot at this point is that from previous research that actually we published looking at the uh and using fmri but also other uh neuroscience you know scientific tools like for example thermocount um we ended up understanding or exploring the fact that actually the brain is starting to react to specific or general touch in a kind of uh similar but different way so and these actually creates a kind of a question a research question on where ability of the specific manual therapies approach might intercede with a specific brain bottom-up or top-down uh in an old-style way describing the how the brain works because actually now we know and looking in particular in

the inactivism that uh and an active influence the body the brain actually doesn't work without the with the top down or bottom up and bottom outstream so far but in a in a way we actually we ended up saying that probably uh the interaction with the body so using a kind of uh affordance and sense making so creating action into the the uh into the patient and through the action of the uh the therapeutic alliance and therefore through the uh the uh professional and the operator might create a kind of a sharing scenario or a sharing environment that when with the with the concept of predictive processing and active inference might also be explained through the markov blankets communication and markov blankets connection between the uh the the two the two elements uh so the patient and the operator that some something might happen not only through the specific activity that any type of manual therapist can do but also in within a framework of closeness between the um the the two uh the two elements indeed one of the of the early study that also lucas mentioned what actually we did was just touching uh the ankles of um and a healthy patient lay down on the scan and we just change our uh mental attitude on focusing towards the hand or avoiding the attention towards the hand so like mimicking the way in which a um a manual therapist was uh paying attention to what he or she is doing manual manual touch or just being absolutely uh away from the attention of the manuals and therefore in the specific elements of the uh of the of the research was just counting a random bit through the through the headphones so and what actually we we saw is that in in those people that were touched uh through an attention towards the hand there there was an higher activation of the insula and the infra frontal gyrus like in a way the the the perception of the patient is is before the fact that actually is conscious of something happening from an interceptive point of view within the body so and the body is saying do you know what i'm feeling something that actually i'm processing through an interceptive framework in the in the insula and at the same time i'm changing my endos endogenous or exogenous attention based on the way in which you touch me so and therefore this creates a more complex environment in which actually we we we would like to understand better and we think that and the inactive uh approach is one of the and with all its let's say subsections and elements that might be an inter interesting way of interpreting or at least trying to read what it's happening in the complex clinical environment thank you i'll make an observation and then ask a question so robert i really like what you said about uh being motivated from earlier on by what you saw the outcomes of treatment were and then that was kind of alluded to in what we just heard like um you could be touching someone's ankle but it's not just the ankle that might change maybe it's their knee but maybe it's something in their head but then maybe it's

something in their organ or their brain or their mind that idea of like entering at the anomaly is it's very clinical like i'm not that kind of a doctor but that's kind of what i understand to be done like looking for patterns and anomalies and then it's also very at home in the realm of participatory sense making like you also address but also that's kind of active inference which is relative to a generative model of the world observations are either consistent or surprising in some way and so that was just kind of it's an interesting tie-in and i think it just speaks to how even in a qualitative or just a conceptual level across different domains we can like talk about active inference as well as other areas and there are concordances and it's not like one has to nest over the other there are different domains and um approaches so um robert any thoughts on that or i have other questions yeah i think um there needs to be some some kind of a dressing of you got to you know you have a relationship between a practitioner and the patient and i think the participatory sense making can start to somehow dissolve a little bit of the the kind of normative language which has become problematic so so anomalies in the past have been dismissed as in oh you're not getting better because you're not reacting to the treatment properly whereas these types of approaches are saying oh oh there is a shared narrative and i think the narrative medicine movement is also quite important in in these ideas obviously we can't write about that as well but it's it's it's nice i think you've picked up on that sort of multi-dimensional and crossover between different disciplines because i think that's another thing we're trying to make a big point of in we need to reach out to other disciplines and say look we have this way of doing things but we know we're not we don't have the whole picture at all but if we can combine with psychologists psychiatrists i mean all of all of us on the paper we've all gone outside of osteopathy to to look at other disciplines and again that's the kind of another kind of it's like a meta kind of participatory sense-making that goes into i think what we're trying to achieve like someone will look at that figure and you all are saying hey we're the person above the table with the hands on and with medical training but then there's another part that you didn't go into detail on what would be therapy component or the niche modification component so totally agree that the paper is like an invitation and that was really um to my question like we see in other domains and applications of active inference you reframe something that might have been perceived about like um just reward or a normative precision based approach like this is the reference range this you're getting your objective function is to enter that reference range and it's like a red flag if it's not um towards this more perhaps even customized uh reduction of uncertainty um and the body like predicting its own sensation so i just wondered what are the implications does that play a role in how the medical

expert understands the scenario does it play a role in the patient's own understanding of what's happening just beyond the research predictions and experiments that are being described how does this translate into a um clinical experience so one of the things uh it's quite interesting you bringing the you know the concept of nature construction um on the other paper that we have in the pipeline where we explore we actually argue that australia osteopathic care is a kind of a effective inference uh you know an argument around you know the construction of you know a niche which can be a diet uh in the case in most adult interactions in clinical practice but you can also go beyond that and be it with a triadic kind of relationship for example when you're treating a baby where actually it's not a two-way communication between the practitioner and the baby but it's also with with the parents with the mother and and in fact you know the first person you need to bring you need to synchronize is probably with the mother as well because you know if you if you kind of treat the baby indirectly you get there so i think that there's a lot of sort of alignments a lot of synchrony um to take into account into the therapeutic encounter and you know arguably the the kind of the development of a robust uh throbbing alliance is through that when the two people sort of sharing uh sharing a common niche but also sharing a you know common kind of uh you know mental states and predicting uh in a lot of times without any words uh what the response of the other person is is likely to be so it could be uh for some people may sort of sound a little bit uh more kind of esoteric but if we put it through a lens of you know active inference in activism and so on sense you talking about things that are pre-reflective you're talking about things that's just happened when um even down the realms of at times we hear practitioners or our own experiences practitioners on kind of bodily work then you feel like at times that uh you know the patient is seems to be relaxing because you got some relaxation through your own kind of a body and it's a sort of it could be the esoteric side of things but also you could think that maybe that show that demonstrates some form of alignment maybe that demonstrates an element of synchrony that can be done at such a more physiological level so there's a lot of um you know there's a lot of mileage uh in in my opinion and in exploring clinical practice uh particularly in particular in in situations of with an element of of complexity to explore it through them you know these lenses yeah and also yes yeah and also uh these also in line with the um the most recent theoretical paper just published a few a few months ago by antonio damasio and colleagues uh where actually they they were looking at the complexity through the um interceptive nervous system so what actually they called for the first time ever no intercepted nervous system so it's a system that includes not only the fact that actually we need

to understand what it's going on within our body so the old concept between what's the old concept of crane so how do you feel but it's also goes into the fact that the body actually interacts with the environment and also the uh that beautiful issue on uh neuro or neuroscience uh based on uh interception where actually they try to predict you uh say something new in terms of uh interception where the uh the main hypothesis was that the way in which we create interception that is the base for a complex clinical condition is not only related to the fact that the different brain areas and therefore the introspective nervous system it's creating a kind of interaction between all the autonomic and non-autonomic uh areas but also those areas are important for us for decoding what it's happening outside the body and therefore through a system through a descending primary and secondary descending mechanism that actually they're introducing new variables within uh within the system that uh it allows the system to detect and to understand better the complexity and through this complexity created the appropriate changes that might be helped and for us keeping the non-equilibrium state and therefore all the concept in relation to the predicted cause so what is the how we can intercede with that because actually we are using one element that is touch that in a way might be disturbing the system because actually it's using in particular gentle touch or what is called affective touch is disturbing or interacting with the system and just through those uh anatomy functional pathways that actually goes directly to the insula and so through the ct fibers and therefore we are interacting uh in an ascending way uh with that specific areas that are including the um interceptive nervous system that actually actually are they the one on the part for emerging the uh the elements that we are talking about now and why it's important as just to underline what joe said which uh i found very clever and i support that so it's a way of creating a sort of uh entry point that also in the complex system might intercede with something that is multifactorial within the well within the body that not only includes for example pain the perception of pain and the experience of pain and all the elements that are inside the body but also it creates the bond with something that is external and bonding with external it means that actually you are establishing the therapeutic alliance in terms of the dyadic but also indirectly the triadic condition when actually you treat for example a baby or an early person so when actually there is someone there might be a parent there might be the son the daughter or legal guardian actually are bringing the uh that person into you into the clinical encounter and therefore are part of the uh the general uh treatment uh session so i think it's a way by looking at the neuroscience behind that and the theoretical framework and the practical examples that are clinically found can be clinically found it can be clinically

explored i think there is a nice way also to um uh intercede with the uh with a different lens uh to understand complex situations i would add something to yes please torsten okay i would add something to um what was said by jay telly and basharsh was that i would even go a step further i mean we decided to support luka's work financially because it is part of the psychosomatic asparty and our point of view but it's not only that we want to interact our palpation in reaction to interception but also to body sensations to arousals to emotions to belief systems and from my point of view in this case the osteopath is less a therapist but more a co-regulator by the interaction between both of them the awareness of the patient to his own stimulus in different levels will increase and secondly from the beginning osteopathy was interested in interrelationships and this means that we are already in our own profession in a way interdisciplinary so it means for example that also we could look that peripheral hyperglycamine will lead always to a central hypoglycemia to central uh glucopini even so it means you have no more sugar in your brain so even if you would touch and you do all your psychological stuff but by your metabolic system you create a hypoglycemia in your brain even then whatever you do it will not has a big result so as more fragmented areas in the functioning of the patient's health we can include in our treatment and this means basically which was in the past not so much dominant and more active patient because in research it was show proactivity is one of the most factors to be healthy and basically in an australian treatment the patient is laying passively most of the time in the past so here we have not only looking in research how osmatic treatment can help but even change ospatic treatment in a more post-modern way so where we can help the patient even more with our hands instead what was past treatments done in australic treatment before so actually on that theme of kind of origins and futures of osteopathy i'll read a comment from the live chat and anyone else can write a chat too good morning my dear osteopathic brothers and sisters i'm an american osteopath and must leave for work now but wanted to share a history of the still hildreth hospital in kirksville which was an outgrowth of dr stills original concept wishing to have osteopathic applications to mental health there are still papers papers by still which came out of that institution about effects osteopathy had on mental health conditions and early research on osteopathy and mental health from mel friedman who is actually my father so um any thoughts on that and also just keeping in mind the guest stream is kind of a two-way street people inactive inference who are perhaps hearing about osteopathy for the first time and vice versa just what are some of those principles lucas that you mentioned in your presentation informed by osteopathic principles like what are some of those principles and origins and how

has it changed because it was very interesting what torsten had addressed yeah so i think an important distinction that we have to make is between american osteopaths and osteopaths mostly around the world because um i for myself i'm not a medical doctor so in the us the old the do is a medical doctor in osteopathic uh in the osteopathic field so they are um they can treat mental health conditions and psychiatric disorders and it is regulated that they can do this so in germany for example i can't treat someone with a major depression because my training was not oriented to allow me to treat that so it is important for us to to emphasize that we are um not treating mental disorders on our own but that we try to um address psychological factors in people with physical conditions that are visiting us quite frequently so i think that that is another distinction that i have to make between our colleagues from america and then to the point of of the history of osteopathy we came across a few papers um that have argued for an osteopathic psychiatry i think that was involved in this paper a little bit in the earlier times there has been some work done but the work is as your father has outlined quite old so uh we we try to put some new perspectives uh to this discussion i think it's another thing one thing if i may uh would be useful to to add is you know the origins of australia with undertale still and and some of you know his early students including um john martin little john were uh were primarily based around you know new concepts um you know concepts of you know sort of uh unity self-regulation that were kind of uh relatively new in the late 19th century and and a lot of the work was sort of one one interesting and important concept of the concept of adaptation so you know enabling people to adapt to uh to their day-to-day activities to to adapt and interact with the environment unfortunately that was a little bit loss uh lost over the years and endoscopy and other forms of manual therapy chiropractic for musculoskeletal physiotherapy became a long time based on a kind of very biomechanical model of care you know looking primarily at the test structure the musculoskeletal structures the reason for uh quite quite a lot of the of the clinical problems which you know in fact led to um hoover and american australian 1963 to to write a very important paper that not many people actually read or actually acted on which actually called for osteopathy or osteopathic medicine to be ecological medicine because you know it's not about just the theology it's not about sort of using a pure biomechanical biomedical model and try to find what is the cause of the problem what is the pathology so i think you know the interesting thing over the years and actually that's um you know something you know personally attracted me a lot to this and to these new ideas uh is actually these ideas uh kind of are are very much you know in in a sense aligned with those initial ideas of lost geography where everything kind of started from concept of this

sort of adaptation as a central concept around the free energy principle for example the interactions with the environment central consciousness concepts around inactivism so really to think about the person in that sort of unique ecological niche and how can we as practitioners kind of help that person to you know help with self-regulation etc also taking into account the knowledge we have these days about what is actually the role of the nervous system pretty much about you know in the role of our stages for example uh not so much about some sort of the high order cognitive stuff so i think there's there's a lot to extract from you know these frameworks and and not necessarily try to rewrite the history of osteopathy or other forms of nanotherapy but actually applying things in a in a kind of a critical and you know serious manner to actually sort of move forward and move away from you know from sort of you know ideas that actually no longer make that much sense from for example you know practicing entirely on a biomechanical model of of you know of care i like very much what george was saying and basically it was not only whoever it was already little john and mcconnell who gone in this way what hoover explored more and we are on the way of them basically so if we nowadays look also into genetics which was in the beginning of osteopathy not possible look into epigenetics look into un lifestyle factors and with each factor the success ability of a person to get sick or get health increases and if we include this with our patient we are on the way little john mcconnell and who was opened up for us already in the very early times of osteopathy thank you torsten robert and then anyone else yeah i just don't want to lose sight of the fact that the paper is about recognizing that even though we might not be trained to deal with psychological interventions an awful lot of our patients have psychological problems of some order or another i something like 40 of primary care consultations coexist with mental disturbance so whether we like it or not we're dealing with it and this is a this is a way we can go we're not training people we're not saying osteopaths should become psycho psychotherapists or psychologists but we're saying if we raise awareness of these issues is going to help in our patient care i agree very much what robot was saying and even i would go a step further we can't avoid to interchange with emotional being with the ans being with cognitive levels of a person we only can be a little bit more competent when we treat with our patient in these areas and this doesn't mean that we need to be a psychotherapist in treating patients it's the complex systems perspective that the entity is in the niche and that the mind and body are interrelated inseparable and so any model that for some sort of like um credentialing reason chooses to ignore that is going to be operating in peril so how do we recognize this system's complexity but also the amend immense amount of

information you know you could study just the liver and there would be a lot to learn or just the lungs so how do we pull back and actually have a healthy relationship in this very complex setting so here's a sort of um fun question also staying with the anomaly theme i was wondering if throughout your training or experiences had come across any embodied exercises that you think are really good entry points to the predictive body and interoception so like i was thinking of having one's arms in the door frame and trying to raise them out and then when you step out you know the arms will raise and um i was wondering if there was any experiences that might be that anomaly for someone to start wondering if they were interested to look into more of these approaches that you outline in the paper rather than the signal processing perception framework or some other um non or pre-act imp model so i i think the literature is quite dense with examples on on how the the body influences the mind in these kind of instances um for example going back to to struck who published the study with the pen in between the teeth uh showing that people find comics much more funnier if they had a pen between the teeth because this pen activates the supermajor with this the primary muscle of laughing so there are quite a few examples on how simple manipulations of the body framework influences cognition and emotion but i don't have some specific uh exercises uh to to give to patients um if that answers the question right because also daniel i think it depends from the patient's condition so far so if we take into account what we what we said up to now so it means that uh each pain each patient has got its uh his or her own clinical specific condition so and uh going into a protocol lies for example exercise my ended up with patients having different reactions so and therefore the the use of this framework that consider the complexity of the patient within the uniqueness of uh of shame or hair definitely my uh help us to be to have fantasy or to have imagination during the during the during the the treatment and adapt the different possibilities that might be exercising might be uh treatment procedures it might be techniques and might be approaches into the condition of the patient that is not only the patient but is the the patient within the environment within his or her environment and therefore create the optimal way of what we predict that might be optimal for uh for the patient in order to achieve the uh the outcome that was um that needs to be obtained obtained i think there's uh in fact you know i'm just thinking one exercise that i tend to use and it's nothing that i develop myself but it's used by quite a number of practitioners to you know so to teach patients about pain and and about the sensations they feel is for example to use the analogy to do you know the this so you if you ask a patient if for example in cases of patients with chronic pain fear of movement you identified that for example movement tends to be

quite slow and and and certainly restricted not so much because there's a serious limitation in the movement but because there's a fear of movement they tend to use for example as the professional looking if you make a fist and you try to move your wrist it's going to feel uncomfortable and that's in a sense how how your neck is trying to move against you resisting if you try to move relax you know it's the sensations are completely different so you can actually create if you think about these exercises that are used by you know quite a lot of musculoskeletal practitioners to actually start introducing the the sort of some some of the the sensations that we want uh that might actually violate their generative model and change things a little bit so that's a good way of you know try to teach the patients you know this is how you should feel this is how you should move and not so much like trying to move against resistance which happens if you make a fist and try to you know move your wrist in general i like well it's very much also francesco's commentary because in osteopathy we try really to identify uh individualized approaches so there was a nice research from amy cutty where she changed body postures and measured the cortisol and testosterone outcomes but in research it was not confirmed because it was too it was not adapted to a certain person sometimes it's enough that just a slight extension in two verticals can give a person a different kind of breathing experience in another person it could be that just the face could be a little bit released in some muscles like for example lucas was mentioned so i think the approaches to a patient in an ospatic treatment is very individualized and it's not a general techniques for everybody which could help so good as in individualized approaches in relation to different experiences thanks for all the awesome and diverse answers so i'll ask the question um carefully i guess but standardized interventions whether they're measured in milligrams or educational curriculum lend themselves towards being scientifically tested more easily like a two group experiment people who do or don't have the diagnosis any other number of situations and it was really also just highlighted in what torsten added with this individuated experience and the dyad is unique there's never going to be another one like that in the history of osteopathy with those people in that niche interacting that way just wondered um what insights from all of your different areas of expertise start to help us address like the specificity of the particulars of an encounter but also know what kinds of things should be available for population level use or on average you know are eggs good or bad yeah i think this is a really big problem in osteopathy um and research osteopathic research in general is what kind of control groups are we are we doing so if we if we are treating a patient is a mere touch without intention a good control group or no intervention so there are many problems to the to

researching the person-centered nature of osteopathy but also the manual nature of osteopathy and i think francesco is a good person to speak about this because he recently published a paper on control interventions in osteopathy maybe you want to add something francesca thanks lucas so uh i think the this is a very interesting um question uh daniel um for three reasons so the first one is that we are still within um a drug based research paradigm because actually um during the 1980s some word the and then with the development of evidence-based medicine the majority of the research actually was drug-based research and therefore we can control milligrams as you said we can control the the type of administration and all the elements that actually created the randomized control trial very controlled very specific and so on however uh in during the last 10 years uh all these paradigms actually start to like having some creepy things and in which sense because actually uh it's it's very far away from the from the clinical practice and the clinical practice is based on variability and the variable and our clients and the variability in our clients are important to understand what are the clinical effects of a specific treatment so um indeed during the last 15 years this concept of having super rigid randomized control trial in particular in the intervention group and therefore in the control group actually started to change the sun to be molded in a way of creating something it's much more much closer to the reality therefore they ended up with pragmatic clinical trial and pragmatic clinical trial those trials actually are more are closer to the reality in which there is not just one an administration in terms of for example drug treatment but actually they are taking into account all the different elements that might influence the effect of the treatment itself indeed when actually we administer for example a drug but then we looking for long-term effects on long-term outcomes in the intervention group an intervention group then goes back home and they are one group one part of this intervention group is very very happy because they they just got in a new child uh they just had and they just won the lottery and all the positive elements obviously the long-term effects of the the drug treatment is definitely better than the uh the someone instead who for example had lost so what is the the what is the concept underneath that is the concept of the the research is changing and also the methodological elements underneath research are changing now there are pragmatic trials but they're also what they call the benchmarking control trial so there are actually are trials that are important for detecting what is happening during the clinical routine life within the hospital and then adding another intervention to the normal routine care then you see what is the difference between the two pathways so what it's saying by that it's saying that it is true that it's important to control all the variables within a given

intervention but at the same time if we look at the cause effect and therefore efficacy yes it's important if we want to look at what is in instead it's called effectiveness so the clinical effect of the specific treatment then what you do what you specifically do a specific treatment or uh 10 grams 10 grams more or 15 grams left actually doesn't make so that difference because of the what we are talking here and also because probably those 15 grams are less are and probably will have more powerful effects if i'm talking to the patient with the right tone and therefore creating an inducing and increased effect uh based on the uh interaction the narrative the shared narratives that we are talking so in a way it's true but at the same time uh the the recent evidence are shifting the uh the methodological research com concept towards complex intervention thanks for the awesome response and indeed it is mirroring trends in other scientific areas to study um targeted perturbations and measuring dynamical systems after perturbation or testing different dynamical models rather than just summary statistic and t-test and then you know rubber-stamped that so it is absolutely really um interesting um does anyone else have any uh comments or questions or things here please robert i mean there's quite a tradition in the nursing world to look at action research which is real life in real life interventions you know what happens at the ground level and i think that's really where we are that's what we're saying you know what actually happens in the treatment room not what we think what happens but actually what happens and this is a conceptual framework that enables that so almost within active it'd be like the difference between what we think is going to happen like well if the two comparisons of the two different amounts had this effect on the biomarker well then the one that had the better biomarker effect should be used in the clinic right and so that's like staying within the generative model but then always being open to having new information and being able to have um discernment to know whether it's consistent or inconsistent with our expectations but our expectations for complex systems even people we know very well like they can still surprise you and so um but always keeping that openness to even the tactile which is something that a lot of other medical modalities don't do so maybe the tactile five percent of the time it diverts a cohort of patients towards another diagnosis like the tissue is warm but that at the whole population level wouldn't even be statistically significant but it might be very important for a small subset so it's very interesting about like kind of pointing without um dwelling on some of the limitations of the current medical system which i think most people can agree is uh not realizing our preferences and then here's actually a science-informed way when we can start to look at complex interventions in human systems absolutely and therefore there there are still they are starting

we are starting to have also some theories uh behind that uh that can be um a complexity or living system that actually can be also applied into uh this field um this changing view of the research field that up to now as we said was mainly based on was biased towards the former pharmaceutical standpoint any other thoughts or comments on that yeah this is a very big point francesco is saying and all his uh his extremely engagement in in helping us to get more science is incredibly important basically to have a little bit balanced we will never have a full balance because with our treatment we can't make so much money like farmer concern can do but at least uh we have opportunities for uh opening uh doors in in a way that is closer to reality and creating a a way that up to now is the mainstream because actually also the um uh drug based uh treatment actually are uh facing uh big challenges up to now and the more you go either you go in ultra precise medicine because actually we are all going all the way down and deep into precise medicine or you need to go on the opposite side in the middle everything is already discovered so uh therefore there are chances to integrate the two because there are chances because actually we are living in a way that actually we need precise medicine uh but at the same time we need also um complex intervention we need complex way of looking uh in a different way not in a super segmented way uh but in a way that actually it brings everything together i think another piece the puzzle was brought up when someone mentioned the early paper about osteopathic approaches as ecological medicine because in my own research with insect ecology that was always the question well you could have the insects in the lab and then you know what age they were and what they ate but then the temperature and the humidity aren't what they have evolved to experience and you don't know whether they're healthy but then out in the wild they're doing interesting stuff in the way that it's meant to happen but you don't know how old the animal is or you don't know what its physiological condition is or you don't get the measurements and so there's a dialogue there between the more controlled laboratory side which is like kind of precision medicine i mean some people may want to live in a laboratory if that means that they could make more measurements about their body and then there's sort of the field ecology natural history angle that's very um interesting how there's like a medical along with a mental health and ecological synthesis happening in a sense through osteopathy now a lot of different perspectives are helping to combine typified and lucas's very dense and informative slides is one of the ease we mentioned there are many e's and like maybe the term inactive might not come in also it's another heat okay seven e more used um but it also shows how different um ways of describing across fields can prevent their synthesis like again osteopathy might not have used

the term i mean looking back on it isn't it it's almost it's it's not even to say it's definitional it's like an understatement so then there's a lot of latent possibility and then how does that get applied and how does it um change the way that the art and the science are carried out are enacted themselves if anyone else would like to ask a question or like what would you say to those who are learning active inference and just hearing about what you've described for the first time or those maybe from the medical or osteopathic community who are learning about active inference for the first time in sort of closing i would say that provides um a way in which we can actually simplify things you know rather than making them more complex i think there's a lot to you know people would feel scared there's another model and there's another kind of complex thing to look into it in fact if you look at sort of the basic foundations of active inference is actually things make perfect sense and in fact you can simplify uh your interaction with your patients in and become a much more effective lawyer you know the kind of uh you know when you get to that point where you know that things are working well since you know your internal states are active states and and vice versa when i can predict things i think you can can help us with that can also help us to um to understand you know complexity that patients bring to the table and you know the paper is about mental health and oxyoptine you know a lot of people would argue post job is primarily based on focused on musculoskeletal care but as we've been talking about you know most musculoskeletal care and the most musculoskeletal conditions have an element of mental health in a sense the patient that can't get out don't do their normal day-to-day activities when their world stops making sense is likely to become depressed so if you by working on their you know musculoskeletal problems make them feel better indirectly or even directly you are treating their mental health conditions i think can help us to see the bigger picture rather than simply the dysfunction in the body for the brain in fact i definitely echo that with george and i think what i've learned and i've been working in this field for 30 years is this is giving us a handle on some of the uncertainty that goes on as a practitioner a lot of my students get really scared about how to deal with unpredictable things that occur in practice and it can be you know pretty challenging because there's a lot of complexity and i think the big take-home message i'd say is you don't have to be the expert on your patient this gives us a handle yeah i mean you have a hard enough time to be an expert on yourself but to be an expert with another human being is pretty damn difficult this gives us a model framework it's some way it's reinventing things that may other people have said in the past but it's it's starting to bring it all together into a package that says right there's a ecological niche or you could say a shared narrative there's a way of talking

to a patient that you don't have to tell them what to do that we can kind of work out we can facilitate healing and that actually is what initially got me into osteopathy i think this is a very interesting point robert and i it resonates with a thing that is an osteopathic principle that we try to activate the self-healing powers of patients and this implies for me that we are not the therapist who brings up healing in the patient but the patient in itself is generating the healing and therefore we are not the specialists who have to understand the complexity of the patient but to be there as a as a counterpart and evolve with the patient to generate a more healthy life i can't help but remark on robert what you said there about not having to be an expert it reminded me of how like the arm doesn't have to be a theoretical coffee cup subject matter expert it has to implement a strategy of which a broad range are going to be possible and none of them can be pre-stated but it has to implement a strategy to get the coffee cup and so it's just a very interesting way to tie together the trans-disciplinarity with also one of the threads of inactivism and ecological psychology and active inference which is that we're doing and thinking and doing are together rather than waiting to be subject matters on experts on everything before we are able to act in a complex niche that includes us as a part but we're not going to get the whole picture on well this was a very fun discussion it sounds like you all are on a very cool line of research and we really appreciate it in the lab i hope that you'll come back and discuss more as more research occurs thank you thank you daniel thank you very much all right thanks a lot okay have a good day thanks everyone youtube okay bye-bye see you soon